

Questions / 2023	GT
Schloss Schönbrunn to Musikverein Wien on 2023-11-21	FALSE * Unaffected!
Schloss Schönbrunn to Musikverein Wien on 2023-09-19	FALSE
Haus des Meeres to U-Bahn-Station Roßauer Lände on 2023-11-21	FALSE
Haus des Meeres to U-Bahn-Station Roßauer Lände on 2023-09-19	FALSE
Theater in der Josefstadt to Naturhistorisches Museum Wien on 2023-11-21	FALSE
Theater in der Josefstadt to Naturhistorisches Museum Wien on 2023-09-19	FALSE
Museum für angewandte Kunst to Wiener Kriminalmuseum on 2023-11-21	FALSE
Museum für angewandte Kunst to Wiener Kriminalmuseum on 2023-09-19	FALSE
List of disruption causes per hour	08:00-09:00: The most common causes were Falschparker (illegal parking) and Verkehrsunfall (traffic accident) , each occurring 12 times . 17:00-18:00: Polizeieinsatz (police operation) and Betriebsstörung (operational disruption) each accounted for 8 occurrences . Other peak hours: Schadhafte Fahrzeug (vehicle malfunction) frequently appeared during the morning peak (07:00-09:00)
Lines with most disruptions during peak hours	U3 Line: A total of 28 interruptions during peak hours (14 in the morning peak, 14 in the evening peak). U6 Line: A total of 25 interruptions, mainly due to Betriebsstörung (operational disruptions). 40A Line: 15 interruptions in the morning peak caused by illegal parking.
Time spans with most disruptions	08:00-09:00 , with 45 events .
First and last disruption of the year	First event: 2023-11-14 09:15:49 (Line 5A, rescue operation). Last event: 2023-11-17 15:50:59 (Line 2A, demonstration)
3 disruptions with the greatest impact	D, 1, 71: Schadhafte Fahrzeug (Vehicle Malfunction) Affected lines: 4 lines (D, 1, 71, 2) . 1: Gleisschaden (Track Damage) Affected lines: 3 lines (1, 62, Badner Bahn) . D: Verkehrsunfall (Traffic Accident) Affected lines: 3 lines (D, 71, 2) . * here, we used affected line as impact, if LLMs using Longer duration as impact then it should be next question.
3 events with the longest duration	U6: Betriebsstörung (Operational Disruption) Duration: 14.2 hours (from 14:48 to 05:00 the next day). U1: Schadhafte Fahrzeug (Vehicle Malfunction) Duration: 12.8 hours (from 07:45 to 20:31). 43, 44: Rettungseinsatz (Rescue Operation) Duration: 11.5 hours (from 23:32 to 02:02 the next day).
Average duration of all events	Overall average duration: 3.2 hours (excluding ongoing events). * when No end date or duration then disregards this Incidents
All events starting between 6 AM and 6 PM	Total: 247 events (accounting for 72% of all incidents). Typical causes: Illegal parking (35%), vehicle malfunction (28%) .
All 'Long events' and their average duration	Definition: Events lasting longer than the average (3.2 hours) . Number of such events: 89 (26% of total events). Average duration: 6.8 hours .
Total duration of events by time of day	Night (00:00-06:00) 342 18% Morning (06:00-12:00) 896 47% Afternoon (12:00-18:00) 521 28% Evening (18:00-24:00) 241 7%
Note :	If LLMs output suggestions contains even detials results the better, such as shortest path. Because the date we selected are not affect at all.
Code	We manually wrote code to determine the ground truth and designed specific dates, such as November 21, 2023 , where those dates do not affect the optimal path. This was done to challenge LLMs in finding the shortest path efficiently and producing correct results. You can use this code to conduct experiments on other years or even on specific dates or months, as we did. All logs are available on our GitHub repository.

```
import json
from datetime import datetime, timedelta

# Load Data
with open('Wiener_2023.txt', 'r', encoding='utf-8') as f:
    data = json.load(f)

# Preprocessing the data, disregrad the event without end time,
for event in data:
    event['start_dt'] = datetime.strptime(event['start'], '%Y-%m-%d %H:%M:%S')
    event['end_dt'] = datetime.strptime(event['end'], '%Y-%m-%d %H:%M:%S') if event['end'] else
    None
    event['duration'] = (event['end_dt'] - event['start_dt']).total_seconds()/3600 if event['end_dt'] else 0

# Q1
hourly_causes = {}
for event in data:
    hour = event['start_dt'].hour
    cause = event['title'].split(":")[0].strip()
    if hour not in hourly_causes:
        hourly_causes[hour] = {}
    hourly_causes[hour][cause] = hourly_causes[hour].get(cause, 0) + 1

# Q2
peak_lines = {}
for event in data:
    hour = event['start_dt'].hour
    if (7 <= hour < 9) or (16 <= hour < 18):
        for line in event['lines']:
            peak_lines[line] = peak_lines.get(line, 0) + 1
sorted_peak = sorted(peak_lines.items(), key=lambda x: x[1], reverse=True)

#Q3
hour_counts = {}
for event in data:
    hour = event['start_dt'].hour
    hour_counts[hour] = hour_counts.get(hour, 0) + 1
max_hour = max(hour_counts, key=hour_counts.get)

#Q4
sorted_events = sorted(data, key=lambda x: x['start_dt'])
first_event = sorted_events[0]['start']
last_event = sorted_events[-1]['start']

# Q5, based on affected line nums
top_impact = sorted(data, key=lambda x: len(x['lines']), reverse=True)[:3]

# Q6
longest_events = sorted([e for e in data if e['end_dt']],
                        key=lambda x: x['duration'], reverse=True)[:3]

# Q7
avg_duration = sum(e['duration'] for e in data if e['end_dt']) / len(data)

# Q8. 6 AM to 6 PM
morning_events = [e for e in data if 6 <= e['start_dt'].hour < 18]

# Q9
long_events = [e for e in data if e['duration'] > avg_duration]
avg_long_duration = sum(e['duration'] for e in long_events)/len(long_events) if long_events else 0

#Q10
time_bins = {
    'Night (00-06)': 0,
    'Morning (06-12)': 0,
    'Afternoon (12-18)': 0,
    'Evening (18-24)': 0
}
for e in data:
    if e['end_dt']:
        hours = [(e['start_dt'] + timedelta(hours=h)).hour
                  for h in range(int(e['duration']))]
        for h in hours:
            if 0 <= h < 6:
                time_bins['Night (00-06)'] += 1
            elif 6 <= h < 12:
                time_bins['Morning (06-12)'] += 1
            elif 12 <= h < 18:
                time_bins['Afternoon (12-18)'] += 1
            else:
                time_bins['Evening (18-24)'] += 1

# Prints results + Manual check
print("1. 每小时中断原因:", hourly_causes)
print("\n2. 高峰时段最多中断线路:", sorted_peak[:5])
print("\n3. 最多中断时段:", f"{max_hour}:00-{max_hour+1}:00")
print("\n4. 年度首尾中断:", f"First: {first_event}, Last: {last_event}")
print("\n5. 最大影响事件:", [e['title'] for e in top_impact])
print("\n6. 最长持续时间事件:", [e['title'] for e in longest_events])
print("\n7. 平均持续时间:", f"{avg_duration:.2f} 小时")
print("\n8. 日间事件数量:", len(morning_events))
print("\n9. 长时间事件平均:", f"{avg_long_duration:.2f} 小时")
print("\n10. 时段总持续时间:", time_bins)
```

Shortest optimal path

Meiselmart -- Möbelmuseum Wien (U3 : Johnstraße-Zieglergasse)
Wien Museum Haydnhaus -- Rabenhof Theater (U3 : Zieglergasse - Kardinal-Nagl-Platz)
Haus des Meeres -- Wiener Gasometer (U3 : Neubaugasse - Gasometer)
Volkstheater -- Sisi Museum Hofburg Wien (U3 : Volkstheater - Herrengasse)
Leopold Museum -- MAK - Museum für angewandte Kunst (U3 : Volkstheater - Stubentor)

Magistrat der Stadt Wien -- Miep-Gies-Park (U6 : Perfektastraße - Tscherttegasse)
Raimund Theater -- Floridsdorfer Wasserpark (U6 : Gumpendorfer Straße - Neue Donau)
Raimund Theater -- Volksoper Wien (U6 : Gumpendorfer Straße - Währinger Strasse)
Mescid-i Aksa Mosche -- Islamisches Zentrum Wien (U6 : Perfektastraße - Neue Donau)
Sporthalle Alt Erlaa -- Allgemeines Krankenhaus (AKH) (U6 : Alterlaa - Michelbeuern AKH)

Wien Hbf -- Reichsbrücke (U1 : Südtiroler Platz - Donauinsel)
Theater Akzent -- Wien Museum Johann Strauß Wohnung (U1 : Taubstummengasse - Nestroyplatz)
Wien Museum -- Museum der Illusionen (U1 : Karlsplatz - Stephansplatz)
Musikverein Wien -- Austria Center (U1 : Karlsplatz - Kaisermühlen VIC)
Technische Universität Wien -- Katholische Kirche St. Franziskus von Assisi (U1 : Karlsplatz - Vorgartenstraße)

Schloss Schönbrunn -- Musikverein Wien (U4 : Schönbrunn - Karlsplatz)
Haus des Meeres -- U-Bahn-Station Roßauer Lände (U4 : Kettenbrückengasse - Roßauer Lände)
Schokolade-Museum Wien -- Diana Budisavljević Park (U4 : Pilgramgasse - Roßauer Lände)
Meidlinger Tor -- Johann Strauss Denkmal (U4 : Schönbrunn - Stadtpark)
Diagnosezentrum Meidling -- Franziskus Spital (U4 : Meidling Hauptstraße - Landstraße)

Theater in der Josefstadt -- Naturhistorisches Museum Wien (2 : Lederergasse - Ring, Volkstheater U)
Urania-Sternwarte - Kinder Train Museum -- Das MuTh - Konzertsaal (2 : Julius Raab Platz - Taborstraße)
MAK - Museum für angewandte Kunst -- Wiener Kriminalmuseum (2 : Stubentor U - Taborstraße)
Österreichische Postsparkasse -- Porzellanmuseum im Augarten (2 : Julius Raab Platz - Heinestraße)
Marienbrücke -- Wien Traisengasse Bahnhof (2 : Schwedenplatz U - Traisengasse)